PRODUCT REQUIREMENT DOCUMENT

PRD — Live Stock & Options Trading Scanner

PRD-2026 · AIAvenue Agency

Executive Summary

India's equity derivatives segment on the NSE sees over ₹70,000 Cr in daily F&O turnover, driven by 2.5+ million active retail traders. The vast majority of these traders rely on scattered tools — a price chart for technicals, a scanner for volume, Excel for OI data, and Twitter for market sentiment — creating cognitive overload and missed opportunities. This product is a real-time NSE F&O analysis platform that scans all 188 F&O equities through a 7-filter Pro Scanner pipeline (Market Trend, Sector Strength, PDH/PDL Breakout, Dynamic Volume/ATR, VWAP, RSI 14, OI Buildup), an AI Intraday Scanner with a 7-factor scoring engine, and a 4-Phase Smart Scanner that drills from Market Macro → Sector Heatmap → PDH/PDL → Deep Analytics. Additionally, it provides a NIFTY OI tracker with PCR, support/resistance levels, and a multi-factor sentiment engine. All data streams in real time via Socket.io. Target TAM is ₹250 Cr, SAM ₹62.5 Cr, SOM ₹12.5 Cr in the Indian retail trading tools market.

Problem Statement

Core Problem

Retail F&O traders in India operate in an information-asymmetric environment. Institutional traders have Bloomberg Terminals, Reuters Eikon, and dedicated quantitative research teams. Retail traders have TradingView (limited NSE data), broker-provided tools (basic, ad-heavy, 15-min delayed), and WhatsApp groups (noise-heavy, unreliable). The specific pain points cluster around four gaps:

1. **Data Fragmentation** — Price, OI, volume, sector data, market mood — all in different tools 2. **No Contextual Scanning** — Existing scanners show raw numbers; they don't rank or contextualise 3. **Time Pressure** — F&O decisions must be made in seconds; manual cross-referencing is impossible at scale 4. **Pattern Blindness** — Traders miss macro-to-micro connections (e.g., banking sector strength → specific stock PDH breakout)

User Pain Points

Persona	Pain Point	Frequency	Severity
Retail F&O Trader	Spends 45 mins/day manually screening 188 stocks across 3 platforms	Daily	Critical
Swing Trader	Misses breakout setups because he can't watch 40 stocks simultaneously	Intraday	High
Options Writer	Cannot accurately gauge sector-wide OI buildup for correlated hedging	Weekly	High
Trading Educators	Need to demonstrate professional-grade scanning to students but have no tool to show	Per class	Medium
Proprietary Desk	Junior analysts spend 2+ hours pre-market preparing scan reports	Daily	Critical

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User Pain Points

User Research & Personas

Persona 1: Karan Malhotra — Retail F&O Trader

Attribute	Detail
Age	34
Location	Gurugram
Day Job	Senior Software Engineer (trades 2–3 hours daily)
Portfolio	₹12 Lakh capital; trades NIFTY Bank, Reliance, Infy, HDFC Bank options
Tools Used	TradingView charts + Zerodha's Kite (broker) + Sensibull (options analysis) + Twitter screener
Frustration	"I have 4 tabs open. By the time I piece together volume spike + OI buildup + VWAP crossover on a stock, the move is already done. I need one screen that connects the dots."
Need	Unified scanner with multi-factor ranking; real-time alerts; OI + price divergence detection

Persona 1: Karan Malhotra — Retail F&O Trader

Persona 2: Meera Iyengar — Swing Trader / Educator

Persona 3: Ajay Suri — Proprietary Trading Desk Analyst

Market Analysis

Total Addressable Market (TAM)

Indian retail trading tools and analytics market (2026): **₹500 Cr**

- 2.5 million active F&O traders
- Average annual spend on tools: ₹2,000–₹10,000/trader
- Plus institutional/prop desk spend

F&O-specific tools segment: **50%** of the above.

TAM = ₹500 Cr × 50% = ₹250 Cr

Serviceable Addressable Market (SAM)

Traders who actively use at least 2 paid tools (TradingView + broker/platform) and trade at least 10 lots/month. Estimated **25%** of the active F&O base.

SAM = ₹250 Cr × 25% = ₹62.5 Cr

Serviceable Obtainable Market (SOM)

Target 10,000 paid users + 50 prop desk/institution seats over 3 years:

- Retail: 10,000 users × ₹6,000/year = ₹6 Cr
- Institutional: 50 seats × ₹1,00,000/year = ₹0.5 Cr
- Base Year 3 → scale to **₹12.5 Cr** by Year 5

SOM = ₹12.5 Cr (5-year target)

Market Rationale

- SEBI has tightened F&O regulations (2025–26), increasing the need for professional-grade tools
- Zerodha's trading volume dropped 30% post-SEBI circular; traders need an edge more than ever
- No web-based scanner exists that combines price + OI + VWAP + RSI + volume + sector + market macro in a single view
- TradingView's NSE data is not real-time (15-min delayed for free; real-time costs extra + has gaps)
- The market is fragmented across dozens of small tools; no clear winner has emerged

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Competitive Analysis

Competitor	Type	Strengths	Weaknesses	Our Edge
TradingView	Charting + Scanning	Best-in-class charting; strong community; Pine Script customisation	15-min delayed NSE data (free); no OI data; no sector heatmap; no F&O-specific filters	Real-time NSE data; native F&O OI; 7-filter Pro Scanner with F&O context
Sensibull	Options Analysis	Excellent options payoff diagrams; strategy builder	No equity scanning; no multi-factor ranking; no real-time scanner	Equity + options scanning with factor-based ranking
Amibroker	Scanning + Backtesting	Most powerful custom scanning (AFL); massive library of scans	Desktop-only (no web/mobile); dated UI; learning curve	Web-based modern UI; zero-coding scanner builder; 4-phase guided pipeline
Opstra	Options Analytics	Good OI analytics; Indian market focused	Limited to options chain data; no price-action scanning; no alerts	Full price + OI + volume + technical scanning; real-time alerts
Investing.com	Financial Data Portal	Free; broad coverage; economic calendar	Not real-time; no F&O-specific tools; ad-heavy; no custom scanning	Real-time F&O-focused scanning; ad-free; customisable
Chartink	Indian Stock Screener	Free; Indian stocks; basic technical scans	Intraday data gaps; no OI data; basic UI; limited filters	7 professional-grade filters; OI + sector + market macro; modern UI

Key Differentiators:

- **7-Filter Pro Scanner** — Market Trend → Sector Strength → PDH/PDL Breakout → Dynamic Volume/ATR → VWAP Check → RSI 14 → OI Buildup
- **AI Intraday Scanner** — 7-factor scoring engine that ranks all 188 F&O stocks in real time
- **4-Phase Smart Scanner** — Macro → Sector → PDH/PDL → Deep Analytics drill-down
- **NIFTY OI Tracker** — PCR + support/resistance + multi-factor sentiment engine
- **Socket.io** — True real-time streaming (no polling)
- **Modern Web UI** — Dark theme, responsive, saved scan configurations

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Solution Overview

High-Level Architecture

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│ Frontend (Next.js / React) │
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│ Pro │ AI │
4-Phase │ NIFTY OI │
│ Scanner │ Scanner │ Scanner │ Tracker + PCR │
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│ Alerts │ Saved │ Export │ Settings / │
│ (WebSocket)│ Scans │ (CSV/PDF)│ Scan Builder │
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API Layer (Node.js
+ Socket.io) │
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│ Real-Time Engine (Socket.io) — pushed to all clients │
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│ Scan Pipeline (Market Trend → Sector → PDH/PDL → ...
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│ NSE Data │ Historical │ OI Data Aggregator │
Feed (WebSocket) │ Database │ (NSE F&O Bhavcopy) │
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Scanner Pipeline Details

7-Filter Pro Scanner

Filter #	Name	What It Does	Data Source
1	Market Trend	Determines if NIFTY/BANKNIFTY is trending up/down/sideways	NSE Spot Index
2	Sector Strength	Ranks sectors by composite momentum	NSE Sector Indices
3	PDH/PDL Breakout	Stocks breaking previous day high/low with volume confirmation	1-min bars
4	Dynamic Volume/ATR	Volume > 1.5× 20-day avg AND ATR > 14-day ATR average	Tick data
5	VWAP Check	Price vs. VWAP (above = bullish; below = bearish)	Intraday data

6	RSI 14	RSI between 40–70 for continuation; >70/<40 for reversal watch	5-min bars
7	OI Buildup	OI change > 5% with price change in same direction (valid) or divergence	F&O bhavcopy

High-Level Architecture

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│ Pro │ AI │
4-Phase │ NIFTY OI │ │ │ Scanner │ Scanner │ Scanner │ Tracker + PCR │ │
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└──────────────────┘ │ (WebSocket)│ Scans │ (CSV/PDF)│ Scan Builder │ │
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Scanner Pipeline Details

7-Filter Pro Scanner

AI Intraday Scanner — 7-Factor Scoring

Each of the 188 F&O stocks receives a score (0–100) based on weighted factors:

4-Phase Smart Scanner

NIFTY OI Tracker

Feature Deep-Dive — RICE Priority Matrix

Feature	Reach (users/month)	Impact (1–5)	Confidence (%)	Effort (weeks)	RICE Score
7-Filter Pro Scanner	5,000	5.0	90%	6	3,750
Socket.io Real-Time Data	5,000	5.0	95%	4	5,937.5
NIFTY OI Tracker + PCR	3,000	4.5	85%	3	3,825
AI Intraday Scanner (7-factor)	3,000	4.5	70%	8	1,181.3
4-Phase Smart Scanner	2,000	4.0	75%	5	1,200
Sector Heatmap	2,500	3.5	80%	3	2,333.3
Real-Time Alerts (WebSocket)	3,000	3.5	85%	2	4,462.5
Saved Scan Configurations	1,500	3.0	80%	1	3,600
Export to CSV/PDF	1,000	3.0	90%	1	2,700
Custom Formula Scanner	500	4.0	50%	6	166.7
Watchlist Management	2,000	2.5	85%	1	4,250
Mobile Responsive View	500	3.0	60%	3	300
Dark Theme	5,000	2.0	95%	1	9,500

Priority Ranking (Tiered)

Tier 1 — Core Scanner (Weeks 1–6): 1. Dark Theme (9,500) — Trivial; highly valued by traders 2. Socket.io Real-Time Data (5,937.5) — Foundation of the platform 3. Real-Time Alerts (4,462.5) — Critical for actionability 4. Watchlist Management (4,250) — Basic user workflow 5. NIFTY OI Tracker + PCR (3,825) — Key differentiator 6. 7-Filter Pro Scanner (3,750) — Core product

Tier 2 — Intelligence (Weeks 7–12): 7. Saved Scan Configurations (3,600) 8. Export to CSV/PDF (2,700) 9. Sector Heatmap (2,333.3) 10. 4-Phase Smart Scanner (1,200) 11. AI Intraday Scanner (1,181.3)

Tier 3 — Advanced (Weeks 13–16): 12. Mobile Responsive View (300) 13. Custom Formula Scanner (166.7)

Success Metrics

KPI	Baseline	Target (3 months)	Target (12 months)
Daily Active Users (DAU)	0	500	5,000
Scans executed/day	0	2,500	50,000
User retention (D7)	N/A	40%	60%
User retention (D30)	N/A	20%	35%
Paid conversion rate	N/A	8%	12%
Average session duration	N/A	12 min	20 min
Scanner accuracy (backtest of signals vs. actual moves)	N/A	65%	72%
NPS	N/A	40	60

Roadmap

Phase 1 — Live Scanner Foundation (Weeks 1–6)

Week	Deliverable
1–2	NSE real-time data feed integration (WebSocket); Socket.io server setup; basic stock list (188 F&O)
3	7-Filter Pro Scanner pipeline (Filters 1–3: Market Trend, Sector, PDH/PDL)
4	Filters 4–7 (Volume/ATR, VWAP, RSI, OI); NIFTY OI Tracker with PCR
5	Real-time alerts engine; watchlist CRUD
6	Dark theme UI; scan results view; UAT with 10 beta traders

Phase 1 — Live Scanner Foundation (Weeks 1–6)

Go/No-Go: Scan refresh < 1 second; data latency < 500ms; 10 beta users active

Phase 2 — AI Intelligence (Weeks 7–12)

Go/No-Go: AI scanner correlation ≥ 65% with next-15-min price direction

Phase 3 — Scale & Monetise (Weeks 13–20)

Impact Estimate

Quantitative Impact

Metric	Trader Without Tool	Trader With Tool	Value Created
Daily screening time	45 minutes	3 minutes	42 min/day saved
Setups identified per day	1–2	5–8	4× more opportunities
Win rate (self-reported)	~55%	~65% (estimated)	10% improvement
Avg monthly turnover	₹5 Lakh	₹6.5 Lakh	30% more
Missed opportunities (cost)	~₹15,000/month	~₹3,000/month	₹12,000/month saved

Quantitative Impact

Revenue Model

Break-Even

- Development (4-person team, 20 weeks): **₹40 Lakh**
- Infrastructure (NSE data feed, Socket.io servers, Redis, DB): **₹5 Lakh/month**
- Customer acquisition (₹500/paid user × 8,000): **₹40 Lakh**
- Total Year 1 cost: **₹1.4 Cr**
- **Break-even: Month 16** (at 4,000 paid users)

- **Year 3 ARR target: ₹7.4 Cr**

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